



Review

Blockchain envisioned UAV networks: Challenges, solutions, and comparisons

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ABSTRACT

Unmanned Aerial Vehicles (UAV) or drones is a technology with a huge potential for proving different efficient solutions in the smart city. It has been mostly adopted by military, aviation, civil, and commercial sectors for audio and video surveillance. The communication and security challenges of UAV is well mentioned by researchers and organizations (defense) around the world, although many of the challenges remain unresolved (especially for sensitive and consumer-related applications). Moreover, the conventional network systems may not be very efficient in dealing with the dynamic requirements of UAVs. As the UAVs are generally deployed in tough environments and terrains, therefore it is quite essential to provide a strong and secure network. Motivated from these facts, this paper presents an extensive survey on security issues in 5G-enabled UAV networks as per the literature published till the fourth quarter of 2019. It also presents the taxonomy of existing security issues in 5G-enabled UAV networks. Based on the findings from the survey, we present a Blockchain (BC)-based security solution and a summary of research challenges in the integration of BC with 5G-enabled UAV. Then, we present a case study of implementing BC with UAVs to secure industrial applications.

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1. Introduction

Unmanned Aerial Vehicles (UAVs), also known as drones or Flying Robots (FR), are the aerial devices, which are self-programmed and remotely administered via mobile device by an individual or group of people through wireless communication. They are equipped with sensors, processors, and wireless connectivity [1]. Due to their ease of deployment, dynamic configuration, low maintenance cost, high-mobility, and faster response their applicability is increasing in wide range of applications ranging from civilian, military, logistics, remote monitoring, cinematography, agricultural monitoring, search and rescue, and 3D-mapping [2]. Other diverse sectors where UAVs are used is shown in Fig. 1. These are mainly useful in situations which are too monotonous (i.e., agricultural land surveillance) or too dangerous (i.e., border or nuclear plant surveillance) for human beings. UAVs are restricted to limited geographical region due to their energy and visual coverage constraint. A multi-UAV system (more than one UAVs are operated in parallel) can overcome the aforementioned issues of energy and coverage. It also enhances the observation diversity by sensing data from various point of views. Moreover, it also increases the UAV data reliability and fault tolerance. Above mentioned interactive benefits of FR attracts the customers and industries of diverse domains. The demand of drones for industrial and commercial use was approximately \$0.25 BN in 2013, \$19.3 BN in 2019 and projected to be \$45.8 BN by 2025 [3].

Nowadays, UAV communication is the key research area attracting researchers across the globe and targeting coverage scalability, enhanced throughput, and security. It can be of three types: a) *UAV-to-UAV (U2U)*: In multi-UAV system, UAVs can pass or share information among themselves for scalability, reliability, and precision results [5], b) *UAV-to-base station*: a base station can control or direct UAVs through air interface and UAVs can send analyzed data to base station for accurate and effective results, and c) *UAV-to-satellite*: satellites are used

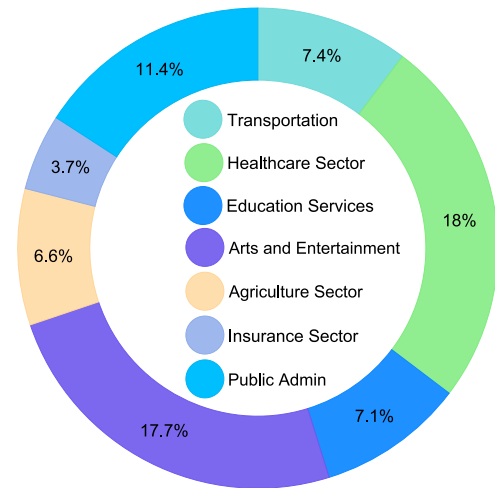


Fig. 1. Day-to-day usage application of UAVs [4].

to track UAVs when UAV leaves the line of sight, then base station shifted to a satellite control.

The security vulnerabilities in UAV communication network were not fully explored by the researchers, but after the UAV hijacking incident of 2009 [6], researchers were started exploring possible cyber-attacks and their countermeasures to make UAVs more secure against jamming, hijacking, eavesdropping, and all other types of network attacks. Over the past few years, the usage of UAVs in different sectors has been increased [3]. According to the report by the Federal Aviation Administration (FAA), more than 100 sightings were happened in a month as shown in Fig. 2. With such an increase in UAV sightings, the transportation departments around the world started creating laws and

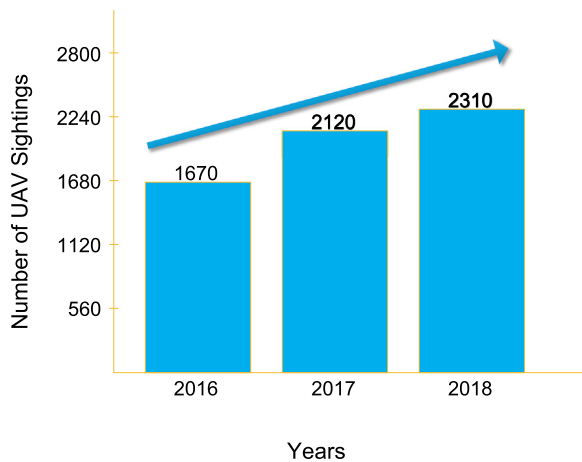


Fig. 2. Year-wise suspicious UAV sightings [7].

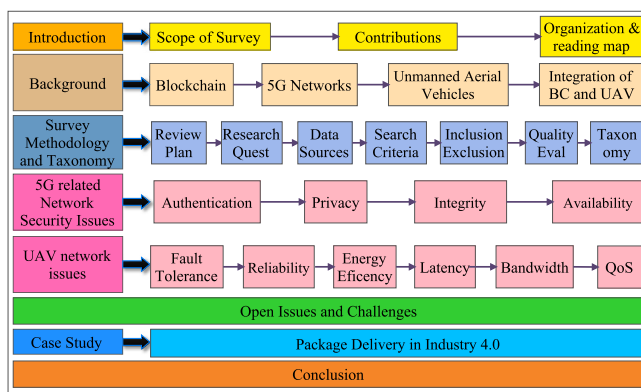


Fig. 3. Structure of the survey article.

impose penalties on UAVs to make sure that usage of UAVs are only in the constructive way [7].

In addition to the security vulnerabilities, the UAV communication network can also be more reliable and has low delay, and fault tolerance [8]. These can be achieved with the usage of fifth generation (5G) communication network, which has already give revolution to the industries, especially to Io-based industries, where delay, energy efficiency, network coverage, and quality of service (QoS) is of prime concern [9,10]. Market research firm IDC predicts that the 5G infrastructure market will grow over \$26 billion by 2023 [11].

Although, 5G network has lots of advantages to UAV networks, but still has its own set of security vulnerabilities like paging occasion and stingrays. Therefore, the 5G-enabled UAV communication network need to be secure against network attacks. To address the aforementioned issues, blockchain (BC) technology is a viable solution with a huge potential [12,13]. It is a chain of blocks (contains transactions), which is connected through the hash value of the previous block in the chain and so on. BC is a distributed ledger in which the stored transactions are immutable, faster access, and transparent to all participating members of the BC [14]. It is secure and reliable because of the decentralized consensus, which makes it suitable for various applications like banking, e-games, music, healthcare, and transportation [15]. Moreover, it has the concept of smart contracts (SC) also called digital agreement written in specific programming language known as Solidity or Go. SCs are self-enforceable programming code (set of rules) between multiple parties involved in a BC [16,17]. It is self-executable, self-verifiable, and tamper proof, which will change the method of interaction in BC and automate various processes like payments, shares, and properties [18].

BC technology enables UAVs to be equipped with cryptographic techniques to ensures secure communication and being used in various applications such as defense and financial applications. It is being adopted by many industries as it reduces the security risks associated with UAV communication. This lead to increased UAV participation in BC network, but current communication channels resist due to latency and scalability issues. So, the 5G as a communication channel in UAV can achieve ultra-low latency (<1 ms), ultra-high reliability (99.999%), and massive connectivity

The integration of 5G and BC technology has a great potential in commercial and defense sectors, especially in military services and must be completely secured. Military information can be confidential information or mission critical and thus should be secured against all possible network attacks [19]. The integration of BC (for security) and 5G (for communication) make the UAV communication more secure against network vulnerabilities. This paper reviews the various issues of 5G-enabled UAV system and tries to find BC-based solutions, which can be better than the traditional ones.

1.1. Scope of survey

Many of the review papers published till date by different authors on the various security aspects of UAV communication network [20–35]. Most of the reviews as per authors knowledge have focused on single or few security issues and not explored the suitability of BC to secure UAV communication. But, there is no exhaustive survey exists that considers all possible security vulnerabilities of UAVs communication system. The proposed survey covers the background of the BC technology, 5G network, UAV communication, and the integration of all three, along with the taxonomy of security issues in 5G-enabled UAV. Jesus et al. [26] presented how BC can be used to resolve security and privacy issues in IoT. Khan et al. [21] and Banerjee et al. [24] reviewed the security, privacy, and other related issues in IoT from the articles published after 2016 and tried to suggest BC-based solutions and research challenges for the same.

But, there are some literatures available, which have not discussed about the integration of BC technology and 5G network. Authors in [22] focused on the cyber-security issues of UAVs network, whereas [31] highlighted many UAV related security and privacy challenges and suggested potential solutions. Later, Gupta et al. [20] focused on identifying UAV communication related issues (like routing and energy efficiency) solutions. Ahmad et al. [27] and Ji et al. [33] described the security challenges and requirements that the 5G technology is facing. Authors in [25] highlighted various authentication and privacy solutions for 4G and 5G networks. Noor et al. [28] analyzed the recent research on IoT security and presented a survey on that, whereas Samaila et al. [29] identified the IoT application domains and highlighted their security requirements.

Ferrag et al. [23] also explored the IoT security issues and challenges in-depth and presented a detailed survey on authentication protocols in IoT. In [30], authors created a generic model for IoT systems while describing the various challenges of IoT. Mohsin et al. [34] created a taxonomy for how BC can be used for authentication in networks. Makhdoom et al. [32] focused on the challenges of adopting BC in IoT and proposed some solutions. Table 1 presents a summary of these surveys and their differences with the proposed survey.

1.2. Contribution of this survey

Presently, several review papers exists in the literature, but as per my knowledge, none of it cover all possible security and privacy issues in UAV communication. Following are the contributions of this paper.

- We analyze the security and privacy issues in UAVs with 5G communication network and present a comprehensive review on it.

Table 1
Comparison between the proposed and existing UAV networks survey.

Authors	Year	Objective	Merits	Demerits	1	2	3	4	5	6
Gupta et al. [20]	2016	To survey work done toward various UAV communication network related issues	Analyses various features of UAV networks, which makes it unique	Focuses on issues of routing, seamless handover and energy efficiency	N	N	Y	N	N	N
Khan et al. [21]	2017	To survey major security problems in IoT and how blockchain can solve them	Maps major issues to solutions in tabular form	Not discussed recent 5G communication trends	N	Y	N	Y	Y	Y
C.G. et al. [22]	2017	To survey cybersecurity literature related to UAVs	Presented a taxonomy for cyber attacks on UAVs	The survey focuses on implications for small UAVs rather than large UAVs	N	N	Y	Y	N	N
Ferrag et al. [23]	2017	To provide a detailed survey of authentication protocols for IoT	Explains over 40 protocols separated into 5 different categories	Only discussed authentication protocols	N	N	N	Y	Y	Y
Banerjee et al. [24]	2017	To survey articles presenting IoT security solutions	Based on the survey, highlights several issues in development of IoT security solutions and how BC may be able to solve them	Only gives conceptual solutions, No experiments to test them	N	Y	N	N	N	N
Ferrag et al. [25]	2017	To survey existing authentication and privacy solutions for 4G and 5G networks	Classification and tabulation of issues and their countermeasures	Does not focus on all issues like availability and integrity	Y	N	N	Y	Y	Y
Jesus et al. [26]	2018	To show how BC can be applied to resolve security and privacy issues in IoT	Working mechanisms of BC, prevented stalker miner attack	Only reviewed stalker attack, BC solutions are not light weight	N	Y	N	N	N	N
Ahmad et al. [27]	2018	To provide an overview of security challenges for 5G networks and their related technologies	Discussed all security issues and their solutions in tabular form	Not focused on privacy issues, BC technology not explored	Y	N	N	N	Y	N
Noor et al. [28]	2018	To provide an analysis of recent trends in IoT security	Presents tools and simulators used for the research as well	Only surveys papers published between 2016–2018, No discussion on BC and 5G	N	N	N	N	Y	N
Samaila et al. [29]	2018	To survey existing approaches and mechanisms for security and privacy in IoT devices	It classifies 9 application domains for IoT and presents their threat models, system models and security requirements	Not discussed about UAVs, mobility, and reliability	N	N	N	N	Y	Y
Sadique et al. [30]	2018	To discuss IoT security challenges	A generic model for any IoT system has been discussed	Not explored 5G and BC technologies for recent communication trends	N	N	N	N	N	N
Lagkas et al. [31]	2018	To review UAV-related challenges, requirements, and solutions	Focused on UAVs and its security and privacy issues	Only focuses on security and privacy	N	N	Y	N	N	N
Makhdoom et al. [32]	2018	To identify the challenges in how BC can be applied to IoT and some proposed solutions	Focused on security and performance issues, not discussed communication issues	Focuses on IoT in general	N	Y	N	N	Y	N
Ji et al. [33]	2018	To analyze 5G security requirements	Analyses security networks of network architecture	Only focused on 5G	Y	N	N	N	N	N
Mohsin et al. [34]	2018	To map how BC technology can be used for authentication	Creates a taxonomy for BC for authentication in networks	Only focused on authentication issues	N	Y	N	Y	Y	N
Mozaffari et al. [35]	2019	Potential benefits and applications of UAVs are discussed	Gives the gist of each section after its completion	Not discussed security and privacy issues	Y	N	Y	N	N	Y
Jensen et al. [36]	2019	To explore how blockchain can be applied to defend against various cyber attacks	Expands on how blockchain can be applied for UAV swarm networks	Only explores one framework to secure a UAV swarm	N	Y	Y	N	N	N
Rana et al. [37]	2019	To improve security of UAVs using blockchain technology	Provides a generic and easy-to-manage access control system.	Does not compare with other solutions.	N	Y	Y	N	Y	N
Aggarwal et al. [38]	2019	To provide a secure and efficient system model for data dissemination.	Compares devised system model with existing schemes.	Does not provide a taxonomy.	N	Y	Y	N	Y	N
Sengupta et al. [39]	2019	To classify attacks and discuss countermeasures for IIoT.	Focused on BC solutions for vulnerabilities.	Does not focus exclusively on UAV networks.	N	Y	N	Y	Y	Y
Proposed Survey	2020	To integrate BC in 5G-enabled UAV network to resolve security, privacy, and communication issues	Presents the taxonomy on security issues in UAV networks	–	Y	Y	Y	Y	Y	Y

1: 5G-focused, 2: Blockchain-focused, 3: UAV-focused, 4: Taxonomy available, 5: Comparative Analysis with other approaches using tables, 6: Categorization of approaches.

- We introduce an in-depth survey of the research challenges in the integration of BC with futuristic UAV communication network.
- We highlight a number of open issues and propose future research directions toward blockchain-based 5G-enabled UAV networks.

1.3. Organization and reading map

The structure of the survey article is shown in Fig. 3. Table 2 lists all the acronyms used in the paper. The organization of rest of the paper

Table 2
The list of abbreviations.

Acronym	Explanation	Acronym	Explanation
5G	Fifth Generation	LTE	Long Term Evolution
ABE	Attribute-based Encryption	MIM	Man in the Middle
APG	Access Point Group	MN	Mobile Nodes
BAA	Blockchain-based Anonymous Access	P2P	Peer-to-Peer
BC	Blockchain	PBFT	practical Byzantine Fault Tolerance
CH	Cluster Head	QoS	Quality of Service
D2D	Device-to-Device	SC	Smart Contract
DACC	Distributed Access in the Cloud	SINR	Signal-to-Interference and Noise Ratio
DoS	Denial of Service	SIPRI	Stockholm International Peace Research Institute
FAA	Federal Aviation Administration	SLAM	Simultaneous Location and Mapping
FR	Flying Robot	U2U	UAV-to-UAV
GCS	Ground Control System	UAS	Unmanned Aircraft System
IoT	Internet of Things	UAV	Unmanned Aerial Vehicle
LoRa	Long Range	URLLC	Ultra Reliable Low Latency Communication

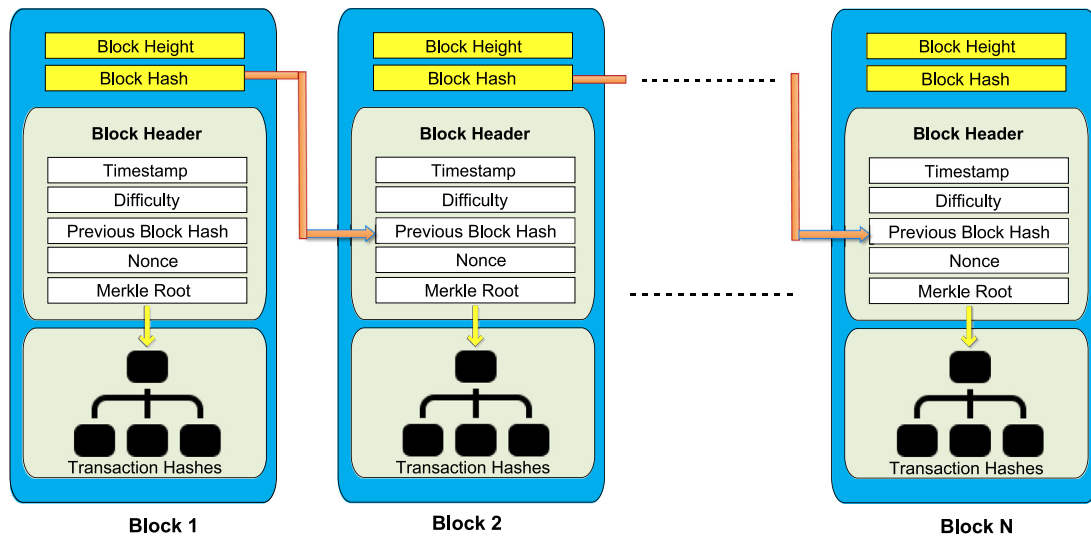


Fig. 4. Structure of a block in the Blockchain [40,41].

is as follows. Section 2 provides the background and history of BC, 5G, and UAV networks. Section 3 presents a taxonomy of the security issues in 5G and network issues in UAV networks. In Section 4, we discuss the research challenges in deployment of BC in 5G-enabled UAV networks. In Section 5, a case study has been elaborated, and finally, Section 6 concludes the paper.

2. Background

This section discusses the background and history of BC, 5G, and UAVs. We further discuss how 5G affects UAV networks and how BC can secure the 5G-enabled UAV networks.

2.1. Blockchain technology

It is a highly disruptive technology coined by Satoshi Nakamoto in 2008 via the usage and benefits of Bitcoins [14]. It is a chain of blocks connected through the hash value of previous block and so on. A block stores the information like block header, time-stamp, block index, block hash, previous block hash, merkle root, and transactions. The structure of a block in a BC is shown in Fig. 4. In BC, the authority is decentralized, which means that every member of the BC network verifies the transaction. In such a peer-to-peer (P2P) network, every transaction is cryptographically marked and validation is performed by checking the transaction with every mining node in the entire network [42]. Every mining node has the copy of whole ledger (i.e., BC), which has immutable records of the transactions occurred between the various parties of the BC. Here, no record can be altered once

saved on the BC, without causing a change in all the blocks, which came later in the BC [43]. Moreover, it allows detail tracking and tracing of transactions of the block [44]. BC combines the benefits of both P2P and cryptography algorithms to guarantee the validity of agreements [45].

BC is having a nifty feature called SC. It is a programming code written in specific language for different BCs (public, private, and consortium BCs) that can be executed when a particular condition is met. It also eliminates the need of third-party systems for executing the transactions. It reduces the processing time and overhead cost in executing transactions over BC network. If any miner node broadcasts a new block to the decentralized BC network, then each member of the BC has an option to either add a block to their copy of BC or simply ignore it. Consensus algorithm (proof-of-work, proof-of-stake, etc.) checks the majority decisions and act accordingly. In nutshell, the role of BC technology is to replace the centralized database and third-party system requirement with authoritative access control mechanism [46]. It helps to prevent data modification and double spending problem.

2.2. 5G networks

It is the next generation cellular network beyond the 4G system, which has various new service capabilities and characteristics such as ultra-low latency (<1 ms), 10–100x connected devices, ultra-high availability (10^{-7} outage per year), 100% coverage capability, 90% energy reduction, ultra-high reliability ($\approx 99.99999\%$), and 1000x bandwidth per unit area [47]. It supports mobile users as well as device-to-device (D2D) communication and working in a high-frequency wireless

band between 28–60 GHz. It also include Long Term Evolution(LTE) bands and unlicensed frequencies to offer users a huge amount of bandwidth [48].

Various technologies can be used to enhance the performance requirements of 5G are software defined network, massive MIMO, millimeter wave, network slicing, virtualization, and D2D communication [49]. 5G communication system can enhance broadband services with massive IoT [50]. The advantages of 5G can be beneficial for Internet of Things (IoT) implementations. It has potential to explode vertical industries, which enables the creation of wide range of services with different levels of security. Moreover, 4G and 5G is vulnerable to various security threats like sybil attack, spoofing attack, modification attack, and so on. This paper highlights the BC-based schemes to maintain data security of UAV networks with SC and consensus algorithms.

2.3. UAVS

This technology began in the early 1920s and is simply an aerial vehicle without a pilot inside. It is a fundamental part of an unmanned aircraft system (UAS), which consists of a drone, a ground control system (GCS), and a communication network between them for exchange of information. It is a complex system which can fly autonomously using pre-programmed flight plans as well as create new flight plan on the fly using Simultaneous Location and Mapping (SLAM). Nowadays, research in industries and academic on it is growing with applications like peacekeeping, combat, monitoring, and package delivery. The overall goal of these systems is to achieve high throughput and enhanced coverage/range. UAVs are considered as the integral part of 5G with major role in coverage and capacity enhancements.

UAVs come in distinguish sizes. Large UAVs can be used in defense or missions critical applications, whereas small UAVs can either to form swarms or perform regular tasks (best for civilian applications). It is well adapted by Indian market as well. India is at the top of the list of importing drones (22.5% of the world UAV imports) as per the report given by Stockholm International Peace Research Institute (SIPRI) [51]. Cellular networks are perfect to support inter-drone communication to increase safety and efficiency of drone operations. 5G technology provides better connectivity and make such connections more reliable for UAV operations beyond line-of-sight. The low latency and high availability help in drone traffic management to avoid mid-air collisions. The extremely low latency facilitates real-time communication between drones as well as drones and their controllers. Uploading of telemetry can allows each UAV to know about their neighboring UAVs. 5G networks have a very large spectrum, which increases the available bandwidth for UAV network, which allows transfer of the large amount of data collected by sensors like video-feed from the cameras. Another bonus of integrating cellular networks with drones is the Quality of Service(QoS) as needed by the drone for different use cases. If a drone is connected to a network its location can be tracked as well [52,53]. BC with its various features to improve transparency, immutability, and security is the perfect fit to handle UAV security issues. UAVs have to take real-time fast decisions, which can be very risky. Moreover, they have to do proper path planning with obstacle avoidance. In case of a single UAV, it must communicate effectively with the GCS to create a safe path. If the UAVs are working in a swarm then they can communicate among themselves and share information to coordinate. This can be done if the information sent and received is reliable and trusted. BC acts as a distributed ledger through which data can be transacted in a secure manner. Thus, in case of a UAV swarm, the UAVs can act as a peers for a BC network to store verified (using consensus algorithm) information. For UAV-to-UAV network, BC security features can keep the communication secure. Thus, BC can integrated with UAV networks to create a secured network for all UAV applications.

2.4. Integration of blockchain and 5G in UAV network

BC technology enables UAVs to be equipped with cryptographic techniques, which ensures secure communication. This allows organizations to use UAVs in such applications that carries critical information such as defense, healthcare, and financial applications. BC becomes a pioneer technology in UAV industry because it minimizes the security and privacy concerns of UAV communication. This increases the participation of UAVs in BC network. But, traditional communication networks such as 4G and LTE-A can hinder the increased participation due to high latency (<10 ms) and low scalability issues. So, the introduction of 5G in UAV communication can bring ultra-low latency (<1 ms), ultra-high reliability (99.999%), energy efficient, and massive device connectivity.

The integration of 5G and BC technology has a great potential in commercial as well as defense sectors, especially in military services and must be completely secured. Military information can be confidential information or mission critical and thus should be secured against all possible network attacks [19]. The integration of BC (for security) and 5G (for communication) make the UAV communication more secure against network vulnerabilities. This paper reviews the various issues of 5G-enabled UAV system and tries to find BC-based solutions, which can be better than the traditional ones [54].

3. Survey methodologies and taxonomy

In this section, we refer to the methodology used to conduct this study, for instance, our search approach and inclusion standards for the final set of papers. Moreover, we present a detailed taxonomy based on our literature review.

3.1. Review plan

The proposed survey begins with the identification of research questions (RQ), related data sources, database search criteria, inclusion and exclusion criteria, and quality evaluation. We identified relevant publications and studies for the systematic survey. The identified material is first check for quality, then only relevant data is extracted for the proposed survey.

3.2. Research questions

The proposed survey identified the existing literature on security and privacy issues in UAV communication. The identified RQs and their objectives used for the systematic survey are listed in Table 3.

3.3. Data sources

A broad overview is required for a comprehensive survey. We used standard peer-reviewed journal databases (for example IEEEExplore, Science Direct, ACM Digital Library, and Springer link) for searching the existing literature and the electronic data sources recommended in [55,56].

Other resources: Technical books, patents, reports, predicting agencies sites, and online literature related to existing surveys are included.

3.4. Search criteria

In this criteria, search using keywords like “Blockchain for 5G-enabled UAV”, (“Blockchain for UAV” AND “5G-enabled UAV”) and other related keywords as shown in Fig. 5. There exist many research papers in which the search string is not either in title or abstract of the paper, for such cases, a manual search process was executed.

Table 3
Identified research questions and their objectives.

Q. No.	Identified research questions	Objective
RQ.1	What are the Security and privacy issues in UAV communication network?	It aims to explore the security and privacy issues in UAV communication.
RQ.2	Which type of issues (communication or network issues) exist and how to address these issues?	It is expected to explore each issue and address that issue using various existing security algorithms to ensure the security and privacy in UAV communication.
RQ.3	What are the different issues and their solutions in various areas of UAV communication?	It is expected to classify the literature on security and privacy issues and better solution to these issues in various areas of UAV communication.
RQ.4	Discuss various taxonomies and comparative analysis of a various survey of security and privacy issues in UAV communication.	Various taxonomies and comparative analysis based on an existing survey of security and privacy issues, which are used to understand the relative advantages and disadvantages of the system in UAV communication.
RQ.5	Discuss various research challenges in the different areas of UAV communication.	It aims to provide information on open issues and research challenges in various areas of UAV communication.
RQ.6	How this study is applicable in real-life scenario?	It aims to provide the case study of usability of BC in UAV network.

Table 4
Quality hiding questions.

Q. No.	Description of question	Answer
Q1	Does the research paper refer to the security and privacy issues in UAV communication network?	Yes
	The papers add an overview of security and privacy issues in UAV communication network where the word “security and privacy issues” are not being used for UAV, such papers are excluded from the literature survey?	No
Q2	Do the abstract, title, and full text of research paper describe the “security and privacy issues of UAV communication network”?	Yes
	Have the abstract, title, and full text of research paper described the security and privacy issues in UAV communication or its other sub-areas?	No

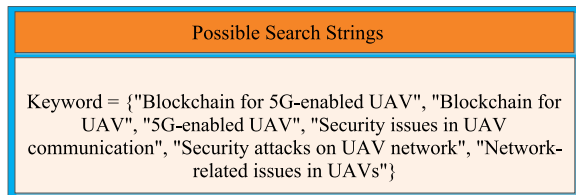


Fig. 5. Search strings.

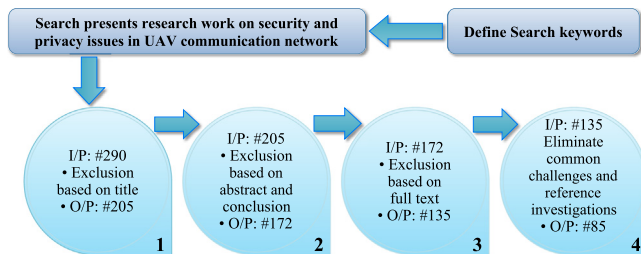


Fig. 6. Criteria of inclusion and exclusion.

3.5. Criteria of inclusion and exclusion

Security and privacy issues available in various application areas, so the search string “security and privacy issues” given irrelevant papers, then filtration was tough at the time of search from the digital libraries. To make a large-scale database, relevant papers of 2019/2020 early access also included. The other survey articles, tutorial papers, technical patents, books and reports, and other resources are included for wider coverage. Fig. 6 shows the stages of inclusion and exclusion of papers. Each stage includes three phases. First is the number of research papers,

second talks about the applied filtration mechanism and third is the number of filtered out research papers.

3.6. Quality evaluation

In this section, quality evaluation has been performed on the papers as the recommendation by the Database of abstracts of reviews of effects (DARE) and Center for reviews and dissemination (CDR) [55]. Quality hiding questions are as given in Table 4, or quality publications have been used to select all research papers.

3.7. Proposed blockchain envisioned UAV networks taxonomy

Fig. 7 shows the taxonomy for 5G and network related security issues in implementing UAV networks. Both affects the performance of UAV system. The detailed description of 5G related security issues is discussed in Section 4, whereas network related issues are discussed in Section 5.

4. 5G related network security issues

Various 5G related security issues in UAV networks are integrity, availability, authentication, and privacy issues. The detailed description about these issues are discussed in subsequent subsections.

4.1. Authentication

Authentication is the process of validating someone (UAV) who declares itself to be. It is important in order to keep the UAV network safe and secure from the malicious users [57,58]. To enhance the security aspects, nowadays two-factor or multi-factor authentications have been preferred. Various authentication factors to be considered are access control, wifi attacks/hijacking, and fabrication attacks, which are explained as follows [59].

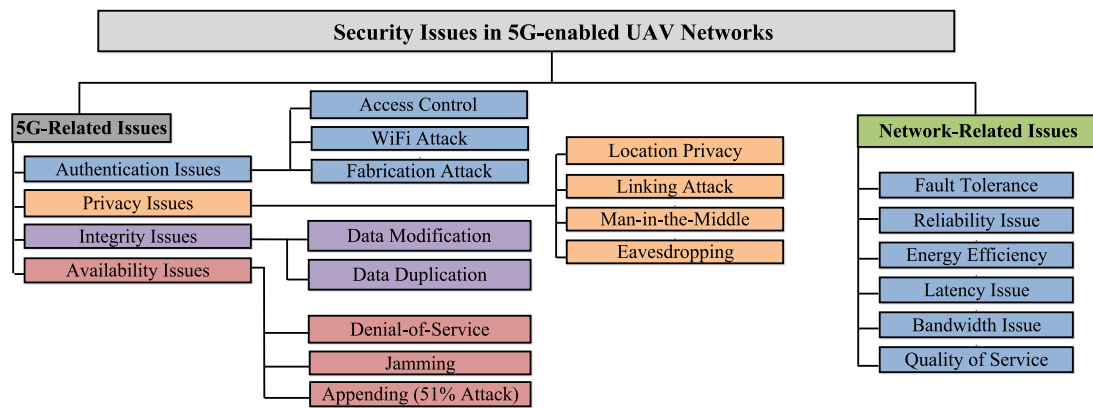


Fig. 7. Proposed Blockchain-envisioned UAV networks solution taxonomy.

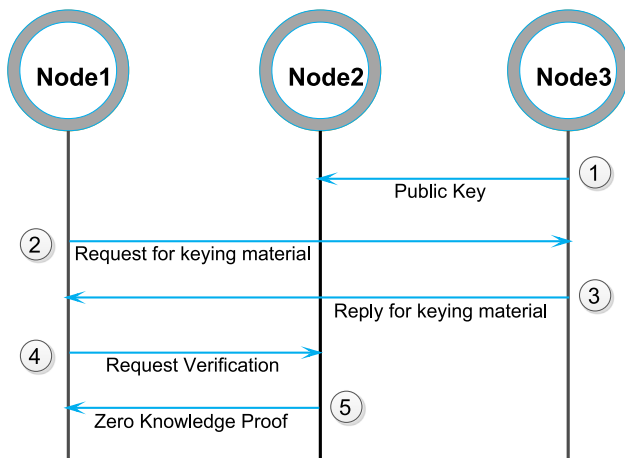


Fig. 8. Procedure of Blockchain-based anonymous access [60].

4.1.1. Access control

It allows only authenticated users to view and use the computing resources. This technique prevents the UAV networks from unauthorized or malicious users to control UAVs or remote data access. It can be either physical (access to limited area) or logical (controls number of connections and data) [61].

Authors [60] proposed a BC-based trusted authentication framework to reduce the operating costs and expenditure of the network. It consists of a BC-based anonymous access (BAA) scheme to address the network authentication with three-way agreement between manufacturer, user, and the operator. BAA handshake process is shown in Fig. 8. Similarly, Rahman et al. [62] proposed a BC-based distributed therapy management system to preserve data privacy and integrity.

Kapitonov et al. [63] designed a method to organize the communication protocols for agents in a swarm that can make the decisions autonomously. It uses decentralized Ethereum BC framework with SC written in Solidity. As BC does not have any 3rd party authentication so there method only allows the user identification via locally generated cryptographic keys. Hence, key distribution mechanism can be used for access control. Authors in [64] proposed a BC-based lightweight system for IoT applications. It eliminates the overhead cost in achieving security and privacy of data. In this, each block is encrypted using Diffie-Hellman algorithm, which ensures, whoever possesses the key is the only one who can access the block data. This shows that key distribution mechanism is one of the strong access control mechanism.

Sharma et al. [65] presented the utilization of UAVs for interoperability between heterogeneous networks to make an ultra-dense wireless network. They have used BC to create trust among UAVs

to reduce the network latency and overheads. The network access security can be ensured with public and private keys. A public key helps in the identification/authentication of drones, whereas private key helps drones to access broadcasted information from other neighboring drones. In continuation of the same, authors of [66] presented used BC as an enabler in the vehicular ad hoc network to ensure network security with the same public-private key concept used by authors in [65]. They have used permissioned BC for reliability, integrity, and validity of data stored in blocks. Moreover, SCs were also used as trusted personal agents to enforce specific privacy rules over common privacy measures [67].

4.1.2. WiFi attack

In this type of attack, a malicious user tries to take control of UAVs by intercepting their data packets over WiFi channels. It is a type of MIM attack which not only captures the data packets, but also can gain access of the UAV system [68]. He et al. [69] examined the cyber-security issues in drone-assisted public safety networks, which can be hijacked through UAV sensors or wireless signals. Authors suggests an attribute-based encryption (ABE) for key management and recommended homomorphic encryption for data aggregation. This approach can protect against WiFi attacks like SkyJet.

4.1.3. Fabrication

It is also known as counterfeiting and used to bypass authentication checks by mimicking information. Also used to access data/service or add new information. Huh et al. [70] presented a platform for IoT applications using BC technology. They have used RSA algorithm where public keys are stored in a BC and private keys are on individual devices. They further claimed that consensus algorithms can also acts as a barrier against forgery attacks. Fabrication attacks can be classified in to message forgery attack and UAV spoofing attack.

Message forgery attack. It is a type of attack, in which a message is sent in such a way as to deceive the receiver about the identity of the sender. Fan et al. [71] proposed a BC-based scheme to solve privacy issues in content-centric 5G mobile networks. It provides high quality privacy protection against message forgery attack via data encryption (light weight symmetric algorithm), access control, and key management. For encryption of data, a symmetric algorithm is used, which is faster and incurs less overhead than attribute-based encryption (ABE) or distributed access control in the cloud (DACC). This technique is an efficient option for UAV-based architectures to secure message communication. In [72], authors explained how BC technology can be used to detect security breaches in swarm robotics. They have used both keys and digital signatures (Fig. 9) for safe transactions in the BC network. Authors in [73] proposed a system to transmit traffic data for UAVs in an efficient and secured manner. It uses a BC-based data transmission mechanism to ensure data security and network robustness against blocking line-of-sight. They have used fingerprints or hash value to authenticate the BC data.

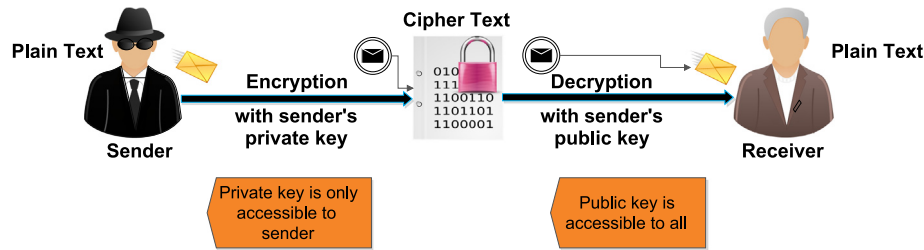


Fig. 9. Digital signature cryptography [72].

UAV Spoofing or malicious intruders. In this category of attack, the malicious UAVs enters into the network silently and try to forge other UAVs credentials. They behave like genuine agents, but actually they are intruders. Ahmadi et al. [74] discussed a way to enhance the resilience of networks using BC technology where they proposed a BC-based system to protect the network against unauthorized intruders or spoofing attempts. Ma et al. [75] proposed a new paradigm of a master-slave BC for pervasive computing, which eliminates the need of high computing ledgers and reduces the transactions execution time. It is suitable for low-powered devices (mobile devices) and consists of a single master node and multiple slave nodes. Master node is responsible for transaction collection and task distribution, whereas slave node is responsible for transaction processing. Moreover, the control model consists of encryption–decryption, key management, and multi-signature authentication for intruder detection.

Dorri et al. [76] presented a lightweight BC-based architecture for IoT, which reduced its overheads in maintaining the security and privacy of data. Similar to [75], it uses multi-signature authentication process with 5G communication, which can increase the transaction processing speed and decrease the latency to feel real-time operations. Later, authors of [75] extended their work and proposed an hierarchical architecture consisting of the IoT devices, local network, overlay network, and cloud [64]. The local network is a local BC which is centrally managed on a single resource capable device. An attacker can try to add a malicious node to the local network, but this is impossible as the list of nodes is only editable by the user. The overlay network is a P2P network where the constituent nodes would be high resource devices. To ensure trust in an overlay BC network, each node maintains a trust rating for the others based on the reputation system. If the trust rating goes below a certain threshold, then we count that node as malicious. The trust mechanism of this architecture is explained in Fig. 10. A relative comparative analysis of various authentication approaches are shown in Table 5.

4.2. Privacy

This section discusses about the state-of-the-art mechanisms to prevent unauthorized users from accessing legitimate information, which is also known as confidentiality [79].

4.2.1. Anonymity and location privacy

Anonymity is to prevent the tracing of own identity, which accessing the web. "If we can prevent others from finding our identity, then others may not be interested in accessing our information". It also prevents network surveillance and traffic analysis. There are many free-ware software available to maintain anonymity such as Tor and freenet. Yang et al. [80] addressed the issue of trusted access with low network cost by an anonymous access identification strategy. In this strategy, the trusted identification is carried out by the consensus among selected parties. To maintain anonymity, a provenance verified public key is generated and to maintain privacy, the data is encrypted and can be accessed using private key. Authors have verified their methodology for privacy and anonymity using zero knowledge proof and congruential method.

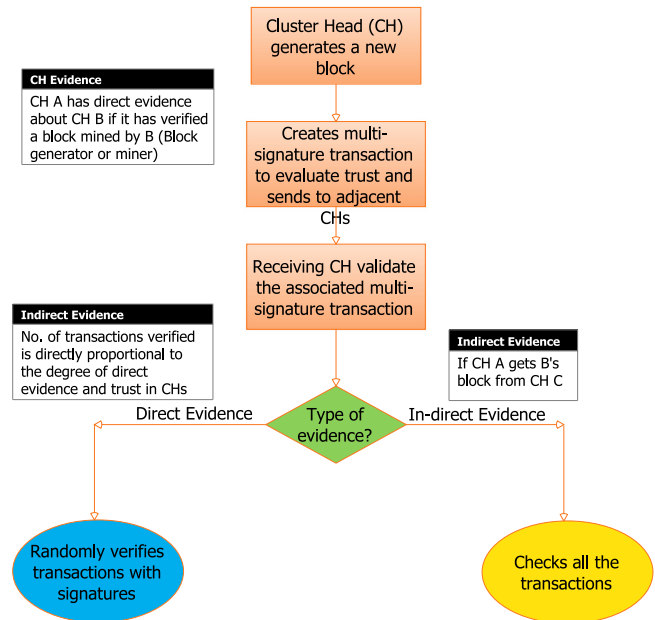


Fig. 10. Trust Mechanism for overlay network [64].

Authors in [73] proposed a BC-based system to transmit UAVs traffic data in an efficient and secured manner. BC ensures data transmission security and makes the network robust against blocking line-of-sight. BC allows data visibility to only their participating members and also ensures anonymity as it allows receiver to identify the sender, but not vice-versa. Later, [81] proposed a BC-enabled architecture for authorization and privacy for mobile platform and services. Its usage is recorded on the ledger as a hash value (unalterable) only after the authentication from the BC members. It is hard for the attacker to identify user from the stored hash value on the BC (anonymity).

4.2.2. Linking attack

In this attack, an attacker links multiple transactions with the same key to find the ID of a neighboring users. Dorri et al. [82] presented the security analysis of their proposed BC-based smart home framework. To maintain the anonymity and preventing linking attack, this framework used a unique key for each device data for their ledgers. They tested it with Cooja simulator and claimed that their proposed framework has very low change in overheads — time (20 ms) and energy (0.07 mj).

4.2.3. Sybil attack

It is a well renowned attack in P2P communication networks where each node uses multiple identities at the same time to undermine the trust reputation of the system. Its main motive is to gain maximum influence in the network to perform illegal actions. It can be either direct or indirect. Examples of such attacks are: (i) multiple fake accounts using same identity, 51% attack in BC, and multiple fake reviews on e-commerce websites by single identity. *How BC restricts*

Table 5
Relative comparison of approaches for Authentication in UAV networks.

Authors	Year	Objective	Approach	Technology used	1	2	3	4
Dorri et al. [64]	2016	To propose a lightweight BC framework that retains most of its benefits	Diffie–Hellman algorithm for encryption, Beta Reputation System	Internet of Things, Overlay BC Network,	Y	Y	Y	N
Azaria et al. [77]	2016	Propose a novel EMR system using BC	Smart contracts, DNS-like mechanism	Ethereum , Smart Contracts	N	Y	Y	N
Sharma et al. [65]	2017	Secure drones being used as inter-service provider nodes via BC	Public and private keys	UAVs, BC, cryptography	Y	N	Y	Y
Yang et al. [60]	2017	Architecture for anonymous and efficient access in cloud radio over fiber network	Three-way agreement	Software Defined Cloud Radio over Fiber Network, Opendaylight Beryllium, Hyperledger Fabric	Y	Y	Y	Y
Huh et al. [70]	2017	To use BC to build an IoT system	RSA public key cryptosystem, consensus algorithm	Ethereum, Internet of Things	Y	Y	Y	N
He et al. [69]	2017	Examination of drone cybersecurity issues.	Attribute-based or homomorphic encryption	Drones	N	N	N	N
Dorri et al. [76]	2017	Propose a lightweight BC based architecture for IoT	Multisig and genesis transactions	Immutable Ledger, Smart Home Manager, Overlay, Cloud Storage	Y	Y	Y	N
Kapitonov et al. [63]	2018	Method to organize communication protocol between agents of a system for decision-making.	Locally generated cryptographic keys	Ethereum, Smart Contracts	Y	Y	Y	N
Ahmadi et al. [74]	2018	How BC can increase resilience of airborne platforms	BC with cryptographic keys	BC, cryptography	N	N	N	N
Ortega et al. [66]	2018	Presents technology enablers to ensure cybersecurity in autonomous vehicles	Permissioned BC + private keys	Content Centric Networking, 5G network slicing	Y	N	Y	N
Rahman et al. [62]	2018	Presents an in-home therapy management framework using IoT and BC	BC and Tor	Mobile Edge Computing, Tor, IoT	Y	Y	Y	N
Chao et al. [73]	2018	To transmit traffic data between small UAVs securely and efficiently	Fingerprint/Hash value for authentication, double-blocks checking	UAVs, biometric	Y	Y	Y	N
Ma et al. [75]	2018	New BC scheme feasible for pervasive computing	Content encryption, key management, multi-signature for intruder detection	Master-Slave, Bitcoin platform	Y	Y	Y	N
Fan et al. [71]	2018	BC to solve privacy issues in content-centric networks for 5G	Symmetric key algorithm	Content Centric Networking, 5G	Y	Y	Y	Y
Ferrer et al. [72]	2019	How BC can be applied to swarm robotics	Key and digital signature cryptography	Drones	N	N	Y	N
Aggarwal et al. [38]	2019	To propose a system model having an Ethereum blockchain to secure drone communication	Encryption/Decryption Keys	Ethereum, Internet of Drones(IoD)	Y	Y	Y	Y
Youssef et al. [78]	2019	To propose a secure system architecture for dam surveillance.	Origin Signature of Data, Auth through UAV provider	Sensor Cloud, UAV cloud	Y	Y	Y	N

1: Architecture/Framework given, 2: Implementation done, 3: Blockchain-focused, 4: Comparative Analysis with existing methods, Y: Yes, and N: No

sybil attack? BC uses consensus algorithm (proof of work) to prove the validity and authenticity of the block to be added into the BC network. Transaction in BC are being verified by all miner nodes and rejected if faulty transaction occurs [83].

Mafakheri et al. [84] used SCs in 5G cellular networks to handle transactions between different operators in a secure way. This paper recommended the use of consortium BC as it is more secure than public BCs and also secure against the Sybil attack. The other ways to prevent sybil attack are: (i) assign different powers to each participating members and (ii) cost to create an identity [85].

4.2.4. Man in the middle attack

In this attack, an attacker monitors or modifies the conversation between the parties. Using this type of attack, an attacker can pass false information or manipulate conversation between the UAVs. It can also be used to give false commands between UAV and its ground station. Garcia-Magarino et al. [86] proposed a BC-based technique to maintain security of UAV network by corroboration of information about the events from different sources. Here, the attacker can be a malicious UAV. This approach monitors the behavior and situations of UAVs like

illegal border crossings. It further uses a secure asymmetric encryption with a pre-shared list of official UAVs to reduce the MIM attack.

Kiyomoto et al. [81] proposed a BC-based architecture to ensure authorization and privacy of mobile services. In this, service information is stored in the BC to ensure identifiability, integrity, and non-repudiation of data. Moreover, it also protects against replay attacks, which is an indication of MIM attack where a valid message is maliciously repeated or delayed indefinitely. Authors have used Byzantine fault tolerant scheme for consensus. Further, a public key certificate and the digital signature of the private key is needed for authentication purpose. If any of these requirements are not met, then it is an indication of something wrong with the message and can help to identify MIM attack.

4.2.5. Eavesdropping

In this attack, an attacker secretly listens the communication between the various parties (UAVs). It is also known for either sniffing or snooping attack and is a passive form of MIM attacks, which do not cause/interfere any network transmission [87]. In this attack, an

attacker targets the unsecured communication links used for data transmission and receiving [88]. Many authors across the world are working on preventing eavesdropping attack. For example, Zhang et al. [89] proposed a privacy-preserving scheme without the pairing of vehicular networks and 5G smart grid slice. This method involves two secret keys, Paillier Cryptosystem, and Hash-then-Homomorphic technique to ensure secure and privacy of V2V communication, which would be difficult to eavesdropped. It can be suitably adapted for UAV networks.

Sharma et al. [90] proposed a delegation scheme for matrix multiplication based on BC system to solve multiple security issues with low energy consumption. This method employs multiple BC ledgers, BC handover key, and One Time Password [91], which can strengthen the security to avert eavesdropping or perpetual sniffing attacks. Authors in [72] explained a way in which BC technology can be applied to maintain security in swarm robotics. In BC encryption scheme, keys and digital signatures are preferred to prove the members identity and transaction safety. Authors claimed that if any message can be eavesdropped, still it can be difficult to decrypt and understood by the intruders. The comparative analysis of various privacy schemes given by different authors are shown in Table 6.

4.3. Integrity

It guarantees that the received data is consistent, accurate (uncorrupted in transit), and can only be accessed or modified by authorized parties. It can be achieved using checksum, error checking, data comparisons, and correction protocols [92]. Attacks on data integrity can be data modification and data duplication attack.

4.3.1. Data alteration/modification

It means when unauthorized parties get access to data which makes the data untrustworthy. Here, a case may occur in which an attacker gets access to data in transit and changes it (similar to MIM attack). To handle this issue, Fan et al. [71] proposed a BC-based privacy preserving scheme for 5G communication network. It consists of a content provider, user community, miners, BC ledger, and cloud storage. In this system, delegated proof of stake is used to ensure network consensus. This can immediately catch the modified or compromised data, which is not verified by the majority of the nodes. Fig. 11 shows the procedure for getting content from content providers. Liang et al. [93] proposed a distributed solution to secure drone communication and data integrity using BC technology and a cloud server. It stores the data on BC network via BC receipt, which makes the data secure and also reduces latency of UAV network. BC prevents the data modification (immutability) and preserves data integrity of UAV network. This method with 5G-communication enables UAV networks to achieve high bandwidth, high reliability, and ultra-low latency for UAV data transfers. Decentralized BC system, SC, and consensus algorithms ensures the UAVs communication is protected from a faulty data source, an external intruder or a compromised node [63,94]. Any changes made to data are recorded as a transaction in the BC, thus maintain the data integrity.

Authors in [65] presented a BC-based trust management scheme to reduce network overheads and latency. In this, all transactions are recorded on the BC and distributed to all participating nodes by the parent/coordinator/controller node, so integrity can be maintained in the system. Authors in [62,65–67,70,75] used BC technology in different contexts to preserve data integrity in related to data modification attack.

4.3.2. Data duplication

In this security issue, the data has redundant copies, which must be eliminated to maintain data consistency and increases network throughput. Motivated from these issues, Ferro et al. [95] suggested a method to use BC to ensure authenticity of information fetched from IoT devices. They used SC, which ensures the data integrity. Uses of

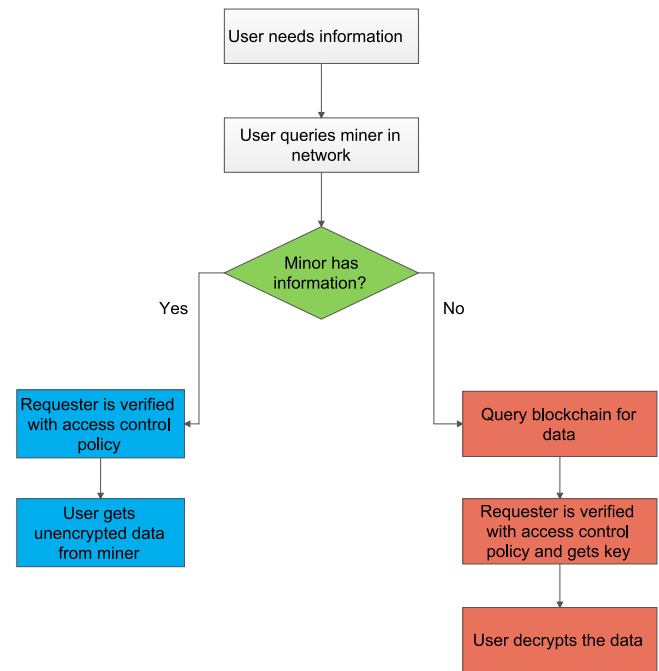


Fig. 11. Procedure for getting content from providers [71].

BC platform further removes duplicate data to make the information more reliable. Dasu et al. [96] discussed modes for air traffic control of drones. To allocate flight space for the time of flight of each zone, decentralized traffic management is an option [97]. Instead of asking a central authority for approval of every drone flight, we use a decentralized BC. For approval, a drone adds a space request to the BC, but only after consensus request is accepted. Consensus can only be established if the respective space is free and nobody else is going to use it. It guarantees that no two drones get the same flight space at the same time so, no data duplication can occur. The comparative analysis of various data integrity schemes given by different authors are shown in Table 7.

4.4. Availability

In availability, the information must be available 24 by 7 to the authentic users of the service. Various attacks on UAV networks against availability of the system are explained in subsequent subsections.

4.4.1. Denial of Service (DoS)

It is an attack in which the attacker tries to overload the server to prevent legitimate users from accessing the services or resources [99]. Bozic et al. [100] presented some applications of BC in network systems. They highlighted the geographical distribution of BC nodes that provide an advantage against single point failure issues. It is difficult to accomplish DoS attacks as the BC ledger is distributed among all participating members. Thus, even if some requested information is not coming from one UAV/node in the network, then it comes from another node. Dasu et al. [96] discussed modes for air traffic control of drones. Instead of asking a central authority for approval of drone flight, they suggested BC-based distributed system. As this approach is decentralized, the lack of central authority would stop denial-of-airspace attacks. Likewise, Kiyomoto et al. [81], Huh et al. [70], and Dorri et al. [64] proposed BC-based approaches against resilience to DoS attack.

Table 6

A relative comparison of approaches used to maintain Privacy in UAV networks.

Authors	Year	Objective	Approach	Technology used	1	2	3	4
Dorri et al. [64]	2016	To propose a lightweight BC framework that retains most of its benefits	Diffie–Hellman algorithm for encryption, Beta Reputation System	Internet of Things, Overlay BC Network,	Y	Y	Y	N
Yang et al. [60]	2017	Architecture for anonymous and efficient access in cloud radio over fiber network	Three-way agreement	Software Defined Cloud Radio over Fiber Network, Opendaylight Beryllium, Hyperledger Fabric,	Y	Y	Y	Y
Kiyomoto et al. [81]	2017	Proposed an architecture for BC based mobile service authorization	parallelization of BC service	–	Y	N	Y	N
Dorri et al. [82]	2017	To show how their BC-based smart-home framework is secure	Shared key	Internet of Things	Y	Y	Y	N
Chao et al. [73]	2018	To transmit traffic data between small UAVs securely and efficiently	Fingerprint/Hash value for authentication, double-blocks checking	Biometrics	Y	Y	Y	N
Mafakheri et al. [84]	2018	To apply BC and smart contracts to cellular mobile networks	Proof of Stake and decentralized Byzantine Fault Tolerance	Smart Contracts	N	N	Y	N
Sharma et al. [90]	2018	Proposal of a secure DMM schema based on BC	One hop communication	Distributed Mobility Management, SDN, Fog Networks	N	Y	Y	N
Zhang et al. [89]	2018	To secure vehicle to vehicle communication	Using secret keys and an aggregation technique	5G smart grid slice, autonomous vehicles	Y	N	N	Y
Garcia-Magarino et al. [86]	2019	Proposed a secure surveillance data transmitted system through UAV networks	Corroboration of information from various sources with asymmetric encryption	UAV simulator, Hyper-ledger	N	Y	Y	N
Ferrer et al. [72]	2019	How BC can be applied to swarm robotics	Key and digital signature cryptography	UAV, BC, cryptography	N	N	Y	N

1: Architecture/Framework given, 2: Implementation done, 3: Blockchain-focused, 4: Comparative Analysis with existing methods.

Table 7

A relative comparison of approaches used to maintain Integrity in UAV networks.

Authors	Year	Objective	Approach	Technology used	1	2	3	4
Liang et al. [93]	2017	To secure drone data collection and communication with BC	Moving data processing to the cloud	Chainpoint, Ubuntu, Apache,	Y	Y	Y	N
Sharma et al. [65]	2017	Secure drones being used as inter-service provider nodes via BC	Public and private keys	N	Y	N	Y	Y
Huh et al. [70]	2017	To use BC to build an IoT system	RSA public key cryptosystem+private keys	Ethereum, Internet of Things	Y	Y	Y	N
Dorri et al. [82]	2017	To show how their BC-based smart-home framework is secure	Shared key	Internet of Things	Y	Y	Y	N
Ferro et al. [95].	2018	To show how BC can be used to maintain authentication of information to be used by an IoT device.	Consult the tamper-proof BC to check integrity of received information	Smart Contracts, Ubuntu, Ethereum	Y	Y	Y	Y
Kapitonov et al. [63]	2018	Method to organize communication protocol between agents of a system for decision-making.	Locally generated cryptographic keys	Ethereum, Smart Contracts	Y	Y	Y	N
Dasu et al. [96]	2018	How air-traffic control can be done for UAVs using geofencing	Consensus	Geofencing, UAVs	N	N	Y	N
Rahman et al. [62]	2018	Presents an inhome therapy management framework using IoT and BC	BC, Tor	Mobile Edge Computing, Tor, Internet of Things	Y	Y	Y	N
Ma et al. [75]	2018	New BC scheme feasible for pervasive computing	Content encryption, key management, multi-signature for intruder detection	Master-Slave, Bitcoin platform	Y	Y	Y	N
Fan et al. [71]	2018	BC to solve privacy issues in content-centric networks for 5G	Local caching	Content Centric Networking, 5G	Y	Y	Y	Y
Youssef et al. [78]	2019	To propose an secure blockchain-based system architecture for dam surveillance.	Permanent Proof	UAV broker, UAV cloud, sensor cloud, Bitcoin	Y	Y	Y	N
Islam et al. [98]	2019	To propose a blockchain-based data acquisition scheme for IoT data.	Merkle Tree	Internet of Things, Mobile Edge Computing	Y	Y	Y	N

1: Architecture/Framework given, 2: Implementation done, 3: Blockchain-focused, 4: Comparative Analysis with existing methods.

Table 8

A relative comparison of approaches used to maintain Availability in UAV networks.

Authors	Year	Objective	Approach	Technology used	1	2	3	4
Dorri et al. [64]	2016	To proposed a lightweight BC framework that retains most of its benefits	Clusters + List of valid public keys	Internet of Things, Overlay BC Network,	Y	Y	Y	N
Bozic et al. [100]	2016	Described how BC can be applied to network systems	Smart Contract	BC	N	N	Y	N
Huh et al. [70]	2017	To use BC to build an IoT system	RSA public key cryptosystem, private keys	Ethereum, Internet of Things	Y	Y	Y	N
Dorri et al. [82]	2017	To show how their BC-based smart-home framework is secure	Shared key	Internet of Things	Y	Y	Y	N
Dorri et al. [76]	2017	Proposed a lightweight BC based architecture for IoT	Trust method — Random validations	Immutable Ledger, Smart Home Manager, Overlay, Cloud Storage	Y	Y	Y	N
Kuzmin et al. [101]	2018	Presented the concept of integration BC to UAV networks	Smart Contracts	BC, UAV	N	N	Y	N
Dasu et al. [96]	2018	Described the air-traffic control for UAVs using geo-fencing	Consensus	Geo-fencing, UAVs	N	N	Y	N
Fan et al. [71]	2018	BC to solve privacy issues in content-centric networks for 5G	Local caching	Content Centric Networking, 5G	Y	Y	Y	Y
Sharma et al. [102]	2019	To present a novel blockchain-based drone-caching scheme for ultra reliability.	Drones as on-demand caching servers	Mobile Edge Computing	Y	Y	Y	N
Youssef et al. [78]	2019	To propose a system architecture for dam surveillance	data anchoring	UAVs, Cloud	Y	Y	Y	N

1: Architecture/Framework given, 2: Implementation done, 3: Blockchain-focused, 4: Comparative Analysis with existing methods.

4.4.2. Jamming

A jamming attack is the transmission of radio signals that disrupt the communications between the parties by decreasing the signal-to-interference and noise ratio (SINR) [103]. Jammers can be used or turn-off the entire system to stop data transmissions temporarily. This can make the system inaccessible from other legitimate users. Jamming attacks are classified as constant (continuous random data generation with some intervals), deceptive (continuous random data generation without intervals), random (random data generation without caring of their energy level), and reactive (intelligent jamming when required) jamming attacks [104].

Kuzmin et al. [101] presented a concept in which each UAV in a swarm is a node in a BC network. It helps to minimize threats to data integrity and privacy, as each drone can have a copy of the BC and thus, can be able to continue on its path even if communication is jammed. The BC can have a SC which allows it to continue its work without communication from other drones in the network.

4.4.3. Appending (51%) attack

It is a potential threat on BC network where more than 50% of the mining hash rate is being controlled by group of miners [105]. Their aim is to resist the confirmation of verification of new transactions occurred in the BC network. But, this attack does not able to gain the full control of the BC network. Dorri et al. [76] showed a lightweight BC based architecture for IoT that reduced its overheads, but maintained the security and privacy of BC technology [106]. This architecture also has a hit-or-miss defense against the appending or 51% attack. Even if an attacker controls over 51% of the nodes, due to the trust method used, the attack may be caught. In the trust method, a random part of the transactions in a received block are always validated depending on the trust level between the two parties. Thus, even if over 51% of the nodes declare a block as valid, there is still a possibility for the false block to be detected. The chances of false block detection decreases as the percentage of controlled nodes increases. The comparative analysis of various data availability schemes is shown in Table 8.

5. UAV network issues

Apart from security attacks discussed in Section 4, UAV system can experience network related issues, which hinders the performance of the UAV network. We have considered various network issues such as fault tolerance, reliability, energy efficiency, packet overhead, latency, bandwidth, QoS, and transparency of the system.

5.1. Fault tolerance

Fault tolerance is the characteristics of a system that allows it to keep working even in the case where some of its nodes gets down [66]. Bozic et al. [100] presented applications of BC network system and highlighted the geographical distribution of BC nodes can provide an advantage against single point of failure issues. The fault tolerance of BC is proved in such a way, even if one or more nodes goes down, then there can be no adverse effect on the network. Moreover, when the node becomes online again, it can be synchronized with the others as if it never went offline. Rahman et al. [62] proposed an in-home therapy management framework with BC-Tor based distributed transactions to preserve data privacy and integrity. BC can eliminate the need of a middleman brings robustness to the security implementation. The distributed nature of BC brings redundancy as there are multiple copies of data with no single point failure.

5.2. Reliability

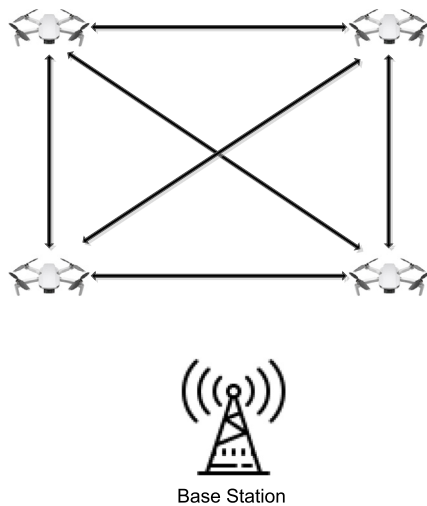
It is the attribute of a system, which shows that it consistently performs according to expectations without degradation. It is also known as quality over time. Yuan et al. [107] gave recommendations for the design of an ultra-reliable communications system for UAV swarms. It focuses on data encapsulation protocols, a lightweight operating system, and a low latency communication protocols for the software protocol stack. To improve the communication system hardware, a new LoRa (Long Range) gateway has been designed to provide support for a local star topology network for UAV swarm. Authors claimed that

Table 9

A relative comparison of approaches used to maintain Reliability in UAV networks.

Authors	Year	Objective	Approach	Technology used	1	2	3	4
Kapitonov et al. [63]	2018	Method to organize communication protocol between agents of a system for decision-making	Locally generated cryptographic keys	Ethereum, Smart Contracts	Y	Y	Y	N
Ahmadi et al. [74]	2018	Presented how BC can increase resilience of airborne platforms	Smart contract, consensus algorithms	–	N	N	N	N
Rahman et al. [62]	2018	Presented an inhome therapy management framework using IoT and BC	BC, Tor	Mobile Edge Computing, Tor, Internet of Things	Y	Y	Y	N
Kaur et al. [108]	2018	To improve data integrity using BC	Via MD5 and multiplicative inverse	Server-Client Interfaces, BC, MD5 encryption	N	Y	Y	Y
Yuan et al. [107]	2018	To propose solutions for UAV swarm communication challenges	Approach is based on software protocol stack and RF hardware	LoRa Radio, UAVs, WiFi, LTE, RF hardware	Y	Y	N	Y
Campion et al. [109]	2018	To propose an architecture to increase UAV swarm reliability	Using cellular network infrastructure to overcome limiting factors of UAVs	ad-hoc network, cellular network infrastructure, 5G, M2M communication	Y	Y	N	Y
Sharma et al. [102]	2019	To propose a blockchain-based drone-caching scheme for ultra reliability.	Drone access points	Mobile Edge Computing, Tor, Internet of Things	Y	Y	Y	N

1: Architecture/Framework given, 2: Implementation done, 3: Blockchain-focused, 4: Comparative Analysis with existing methods.

**Fig. 12.** Mobile network-based UAV Swarm communication [109].

the LoRa performs better than WiFi. But, they have not explored the reliability of 5G communication technology.

Campion et al. [109] proposed an hybrid swarm architecture over cellular network infrastructure as shown in Fig. 12. In this system, a decision making is left to the individual UAVs for high level of autonomy. 5G networks with ultra reliable low latency (URLLC) characteristics proved to be the backbone of UAV swarm architecture. This scheme can enhance the range and reliability of UAV swarm communication network. Authors in [63] developed a communication protocol in swarm communication to make decisions autonomously about their actions. Through decentralized management via BC solves the problem of a security vulnerabilities. As control is individualized, which increases the reliability of nodes/drones with minimal chance of entire swarm failure.

Qian et al. [110] proposed a high-level security management scheme based on BC technology for different IoT devices. A device identification-based key algorithm was used to increase the data security and reliability. Kaur et al. [108] presented a way to improve the security and integrity of existing BC system. They have used MD5, multiplicative inverse, and IP verification to improve the reliability of data and system used. The comparative analysis of various data integrity schemes used in UAV networks are shown in Table 9.

5.3. Energy efficiency

Energy efficiency is a crucial parameter in all the UAV network and also for IoT devices [111]. High energy-efficiency reduces the capital as well as operational expenditures. 5G communication ensures mobile battery life up to 3 days and IoT devices up to 15 years [112]. Many authors across the globe are working on achieving high energy-efficiency. For example, Sharma et al. [113] presented an approach for device discovery in 5G-based IoT and Body Sensor Networks using multiple UAVs. It uses XML charts for device discovery on the basis of network state cost and available energy. This method is 78% more efficient in terms of energy consumption, 21% in terms of end to end delays, and packet loss as compared to traditional solutions. This may help to keep UAVs more aware of their surroundings efficiently. Sharma et al. [65] presented a method with head drone. In this method, the controller/parent/coordinator drone of the BC can change easily as the resources (energy) of the parent drone change. This can make the network more long-lived (energy-efficient) as it will balance the amount of energy left in each drone.

A master–slave BC scheme was designed in [75] for pervasive computing that overcomes the need of heavyweight ledgers and time-consuming transactions. It is well suitable for low-power IoT and mobile devices [114]. Through various experiments, using various mobile devices, the availability, efficiency, performance, and scalability was tested and proved. Sharma et al. [90] proposed a new DMM scheme based on BC to solve various security issues with less power consumption. It can increase energy-efficiency as it decreases the number of addresses to be configured at mobile nodes (MN) as well as number of hops (one hop communication) and associated addresses per transfer.

Dorri et al. [76] presented a lightweight BC based architecture for IoT infrastructure, which reduces its overheads. They implemented distributed trust to validate transactions while reducing the processing overhead at single node. After testing with the NS3 simulator on a network of 50 nodes, it is seen that this architecture decreases processing overhead by 50% due to the distributed trust technique. The comparative analysis of various energy-efficiency schemes used in UAV network suggested by various authors are shown in Table 10.

5.4. Latency

Latency is the time taken by a packet to move from source to destination. Sometimes, round-trip time (source → destination → source) is considered as latency of the network. It mainly depends upon the

Table 10

A relative comparison of approaches used to maintain Energy Efficiency in UAV networks.

Author	Year	Objective	Approach	Technology used	1	2	3	4
Sharma et al. [113]	2017	An energy efficient approach for device discovery	XML charts to perform device discovery based on network state cost and current energy	5G, IoT, UAV, Body Sensor Networks, XML energy chart	Y	Y	N	Y
Liang et al. [93]	2017	To secure drone data collection and communication with BC	Moving data processing to the cloud	Chain point, Ubuntu, Apache,	Y	Y	Y	N
Sharma et al. [65]	2017	Secure drones being used as inter-service provider nodes via BC	Coordinator drone can change as energy decreases	N	Y	N	Y	Y
Dorri et al. [76]	2017	Proposed a lightweight BC based architecture for IoT	BC, SC	Immutable Ledger, Smart Home Manager, Overlay, Cloud Storage, NS3 simulator	Y	Y	Y	N
Ma et al. [75]	2018	New BC scheme feasible for pervasive computing	Master and slave nodes	Master-Slave, Bitcoin platform	Y	Y	Y	N
Mafakheri et al. [84]	2018	To apply BC and smart contracts to cellular mobile networks	Proof of Stake and decentralized Byzantine Fault Tolerance	Smart Contracts	N	N	Y	N
Sharma et al. [90]	2018	Proposal of a secure DMM schema based on BC	One hop communication	Distributed Mobility Management, SDN, Fog Networks	N	Y	Y	N
Xiao et al. [115]	2019	Proposed an iterative algorithm, which optimize each variable block in UAV	Dinkelbach's algorithm	Delay-tolerant decode-and-forward (DF) transmission protocol	N	Y	N	Y

1: Architecture/Framework given, 2: Implementation done, 3: Blockchain-focused, 4: Comparative Analysis with existing methods.

Table 11

A relative comparison of approaches used to maintain Latency in UAV networks.

Authors	Year	Objective	Approach	Technology used	1	2	3	4
Wang et al. [116]	2016	Prototype for adaptive video streaming from UAVs	Content-based compression and video rate adaptation for streaming	Drones, DASH video format	Y	Y	N	Y
Dorri et al. [64]	2016	To propose a lightweight BC framework that retains most of its benefits	Diffie-Hellman algorithm for encryption, Beta Reputation System	Internet of Things, Overlay BC Network,	Y	Y	Y	N
Grasso et al. [117]	2018	To propose a tactile internet architecture for video surveillance systems	Process data on the video capturing system itself	Markov chains, Microcomputer on each drone, Tactile Internet Network	Y	Y	N	N
Dasu et al. [96]	2018	How air-traffic control can be done for UAVs using geofencing	Consensus	Geofencing, UAVs	N	N	Y	N
Rahman et al. [62]	2018	A BC and mobile edge computing based therapy platform	Consensus mechanism by following longest chain rule	IoT, MEC, Tor-BC	Y	Y	Y	N
Pustick et al. [118]	2018	Comparison of distributed ledger protocols from an IoT viewpoint	Combination of off-chain and on-chain protocols	Smart contracts, Ethereum, Hyperledger	Y	N	Y	Y
She et al. [119]	2019	To establish a low-latency framework for UAV communication.	Distributed Antenna system	–	Y	Y	N	N

1: Architecture/Framework given, 2: Implementation done, 3: Blockchain-focused, 4: Comparative Analysis with existing methods.

hardware and network connections used. Grasso et al. [117] presented a 5G-enabled tactile internet-based architecture for video surveillance systems in UAV networks. It aims to transfer captured images with round trip delay of less than 1 ms. To achieve this, the authors installed a microcomputer on each drone. 5G technology can also help decrease the latency. For example, Wang et al. [116] presented an adaptive video streaming algorithms for UAVs to minimize the delay of scheme mentioned in [117]. Authors developed a system called SkyEyes to tackle uncertain wireless link capacities and meet video quality requirements. It uses content-based compression and video adaptation rate based on location sensors and client buffer status to aid their goal. This can change the data size to reduce transfer time.

Pustisek et al. [118] compared the distributed ledger protocols with respect to IoT applications. Through experiments, it has been observed by authors that, BC cannot reliably run on constrained IoT devices. Some BC protocols have been proposed that combine off-chain and on-chain transactions to improve privacy, scalability and latency. Later, authors in [64] proposed a lightweight architecture for IoT applications based on BC technology. It eliminates the processing

overhead, maintains data security, and also privacy benefits. To decrease network overhead and delay, nodes are grouped in clusters. The testing and analysis shows that this multi-tiered BC based architecture has constant performance overhead at best and scales with number of clusters instead of nodes at worst. The comparative analysis of various latency schemes used in UAV network suggested by various authors are shown in Table 11.

5.5. Bandwidth

It is the capacity of a network to transmit maximum data from one point to other via Internet connection. Traditionally, it was measured in bits per second, but nowadays it is measured in millions of bits per second or billions of bits per second due to availability of high capacity links [120]. Various factor hinders its performance such as packet loss, jitter, and latency. Qian et al. [110] proposed a high-level security management scheme based on BC for different IoT devices for their full lifecycle. A BC based cloud platform is suggested with efficient cryptographic algorithm to guarantee low latency and high throughput for data management and security in IoT devices.

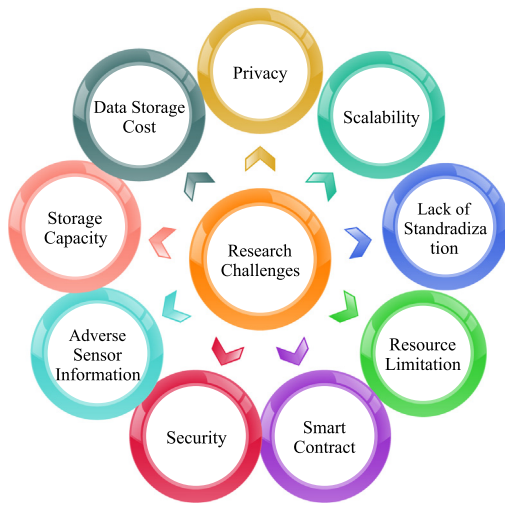


Fig. 13. Open issues and research challenges.

5.6. Quality of service

It refers to the mechanism, which guarantees the network performance by managing network traffic, minimizing latency, reducing packet loss, and reducing jitter. UAV networks need to achieve QoS using standard tools and techniques like jitter buffer and traffic shaping (congestion avoidance and management) [121]. Chen et al. [122] proposed an authentication scheme for 5G ultra-dense networks which enables fast and frequent authentication between the user equipment and access points. In this paper, an algorithm based on optimized byzantine fault tolerance (PBFT) based consensus, trusted access point group (APG) generation, and a reverse screening method is developed to improve the access efficiency. To generate a trusted Access Point group, the PBFT algorithm is modified to also identify trustable nodes, besides verifying the consistency of the information. The scheme only needs mutual authentication between user equipments and node in the APG once. Thus, the user always be in seamless coverage of the APG resulting in better QoS.

6. Research challenges

Although the distributed nature of UAV network provides various advantages in terms of scalability and autonomy, but it bring forward various obstacles creating the need of special attention to the communication platforms also [123]. A number of open issues and research challenges need to be addressed in integrating BC, 5G technology, and UAVs efficiently and effectively. This section discusses the futuristic communication challenges of UAV network with 5G-enabled BC technology to secure communication as shown in Fig. 13. We highlighted some of the crucial challenges such as privacy, scalability, security, SCs, lack of standards and regulations, computing, and resource limitations.

6.1. Privacy

Pervasive sensing systems in UAVs like camera, temperature, gas, and ultrasonic sensors continuously collects data from their surroundings, that can cause unintentional invasions of privacy. The data privacy is a point of concern, as it is required to analyze and predict the data for Guavas path planing. In BC-based UAV system, each UAV has a copy of the distributed ledger, which means some personal or sensitive information is disseminated to all participating UAVs. The UAVs can be able to send enough information to the other UAVs for proper navigation and functioning [124]. Limited information sharing between UAVs in BC-based system is a viable solution, but is a quite challenging problem to be well addressed by the researchers.

6.2. Scalability

It is important in UAVs swarm (group) communication. If the group is small, then, BC platform is well suitable and transactions can be executed with required speed. But, it can be least preferable if groups are big. Motivated from this issue, Pustisek et al. [118] analyzed the transaction execution time as large in BC-based UAV system in case of big UAV swarms. The most popular BC platforms like Bitcoin and Ethereum can perform up to 12 transactions per second. In situations, where small UAV swarms have to communicate with each other in real-time for movement coordination, number of transactions per second can be a research question. Many new platforms are being developed to improve the consensus algorithms and transaction times [125,126], but much more work is needed as UAV swarms are getting bigger day by day due their applicability in almost all area of futuristic smart city.

6.3. Security

The distributed nature of BC has its strengths as well as weaknesses. The benefit of a distributed network is no single point failure. BCs are robust against variety of security threats, but its nature itself make it exposed to a cyber-attack called 51% attack (usually on bitcoins). In this attack, a miner node is controlling over 50% of BC computing power and hash rate. In UAV network, this situation occurs if over half the UAVs in a swarm are hacked. So, a strong robust solution is required in place to strengthen the security of the entire UAV network.

6.4. Smart contracts

SCs are programming code that can be vulnerable to various security attacks due to lack of proper programming practices. Testing SC solutions can be a way forward and specialized tools are also being developed for the same [127–129].

6.5. Lack of standards and regulations

At present, many organizations like IEEE and ITU are working to release new BC technology standards, which helps in BC integration with other technologies, but is yet to be completed [130,131]. There are many active projects going on in BC standards by IEEE to make the BC implementations future-proof, but it require proper guidelines, rules, laws, regulations, and policies. This way, newer BC implementations will remain compatible with the existing ones. So, lot of research need to be done in this area to develop proper technical standards, policies, and regulations for the efficient deployment of BC in UAV networks.

6.6. Resource limitations

UAVs are light weight fly machines with low computation power. But, BC needs high computation power to execute SC and consensus algorithms on UAVs to update all copies of ledgers, which are with other miner nodes. Researchers are finding the ways to design faster and efficient (in terms of power/energy) UAV processors. So, getting maximum operation time from a limited energy resource (battery power) is very crucial [53,132] as energy consumption depends on movement, transmission and internal processing of UAVs. In summation, though researchers are working on the integration of these technologies has a long way to go and a lot of challenges to overcome.

6.7. Adverse sensor information

As the UAVs are equipped with multiple sensors for data gathering from surroundings. There may be a possibility that the sensors can read adverse data from surroundings, which may damage or misguide the UAV.

6.8. Storage Capacity

The storage capacity of BC is still under debate, but in context to UAV communication, UAVs can generate gigabytes of data (audio and video data) in real-time. In this sense, BC may become unsuitable for UAV networks. So, this may be a limitation of the proposed system.

6.9. Data storage cost

Although, BC in UAV communication increases the data security, but also increases its data storage cost. It is because, to store one word (256 bits i.e., 2^5 Bytes) in Ethereum BC requires 20000 Gas. The current Gas price is $\approx 6\text{gwei}$ and the current price of Ethereum cryptocurrency (Eth) is USD 131. So, the final cost to store 1 MB in the Ethereum BC is $\approx$ USD 570 [133].

7. CaSe study

In order to show the implication of BC in 5G-enabled UAV networks, we present a case study on “*Package Delivery in Industry 4.0*”. We also discuss the contextual analysis to show what types of challenges can be faced during its implementation.

7.1. Package delivery in industry 4.0

In Industry 4.0 environment, all precesses are expected to be smart and automated as possible. One of the recent technologies knows as “*UAVs or Drones*” put an huge impact in Industry 4.0. It has applications in almost all sectors of the industry like transportation, manufacturing, fabrication, and package delivery. Commercial drone applications are forecasted to cross the market over \$100 billion by 2020 [134]. Drone II’s application report for 2019 has over 730 use cases from 80+ companies spread over more than 100 countries [135]. Thus, this shows the huge potential of drones in industries.

But, for a particular use case, we are focusing on small UAVs for delivery of packages, food, or other goods from one place to other. Transportation is the fifth largest industry sector for drone applications in 2019. Drone-based package delivery is the area in which authors around the world are working since 2013. Companies like Matternet and Flirtey have been using UAVs for delivery of medicines to rural areas and also for disaster relief supplies. Flirtey has always been partnering with companies like Dominos for air-based food delivery. Established companies like Amazon, Google and Fedex have been looking into UAVs for cheaper post and package deliveries. Though drones may not be much economical for companies than road-based delivery. For regions, where no road infrastructure at all, UAVs will be preferred to deliver over those places. UAVs can transport medicines, goods, and other light-weight supplies to the isolated areas at fraction of cost instantly using drone network compared to building a long road infrastructure.

To implement the above mentioned use case, UAVs must be connected with fast and reliable cellular networks (to connect and control UAVs). Mobile networks are perfect for low-altitude and inter-UAV networks. A mobile spectrum can provide wide area, high quality, and secure connectivity to enhance safety and security of drone communications. With 5G technology, these communications will become more efficient and faster. 5G-enabled drones will have greater bandwidth, lower latency (<1 ms) and higher reliability ($\approx 99.999\%$) [52].

Apart from the benefits of 5G, it also suffers from few drawbacks (security and privacy issues) that still remain unsolved. The main aim of security attack on drone communication is to take control, steal its data, or spoof its positioning sensors mentioned in attack vector as shown in Fig. 14. The attack vector is bifurcated into three layers: (i) data acquisition layer, (ii) UAV control layer, and (iii) monitoring layer. The top-most layer is the data monitoring layer, which has the GCS. It gets data from the drone’s sensors and act accordingly. The

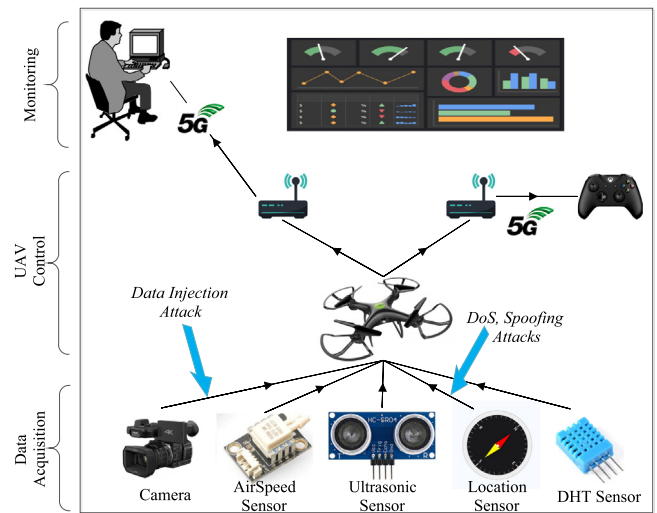


Fig. 14. Attack Vectors in 5G-enabled UAV networks.

bottommost layer has the sensors, which include the location, airspeed, ultrasonic, camera, and temperature sensors. This layer is known as the data acquisition layer. The middle layer is called the UAV control layer, which gets data from the bottom layer and forward it to the GCS and it also executes the instructions received from the GCS. Lastly, this layer also has a remote control for the UAV, which is necessary in case of emergencies like when the autopilot fails or the environment is too complicated to navigate without an experienced pilot and human reflexes. This vector shows the possible attacks on UAVs. Some of the high risk attacks possible in 5G communication are Man-in-the-Middle (MIM), Injection, Sniffing, GPS spoofing, and Denial of Service as shown in Fig. 14 [136–138].

A viable solution to overcome the above mentioned security issues is the integration of BC in UAV networks over 5G communication channel. It has various benefits:

- Data access in BC system is faster as it uses decentralized mechanism along with 5G communication network. (<1 ms latency)
- Data reliability in package delivery would higher as it is being verified by all participating miner nodes (99.999% reliability).
- High data availability in 5G-enabled BC network as is a distributed ledger.
- Fault tolerant network because each miner node has the copy of complete ledger.
- Self-executable digital SCs between the sender and receiver in package delivery eliminates the need of third party systems.
- Package order details cannot be either refused or updated by the receiver as the entry is immutable in nature.
- Resistance from data modification attack (immutability), MIM attack (only participating members can view or add transactions), sybil attack (new miner node can be added after the verification by all participating nodes).

The proposed case study (UAV-based package delivery) system architecture is shown in Fig. 15. It is segregated into three layers as (i) Seller Layer, (ii) Network Layer, and (iii) Buyer Layer. Buyer layer request for products from the seller layer through BC network via Ethereum client over 5G communication channel. A SC in between buyer and seller layers settle the payment before the delivery initiated by the seller layer. Then, seller has initiated the product delivery with data in encrypted form (sender address, destination address, path nodes, and UAV live location). Then, the SC is executed again and verifies the terms and conditions stated between the buyer and seller. After the verification of data, the details are being stored in the BC network. The

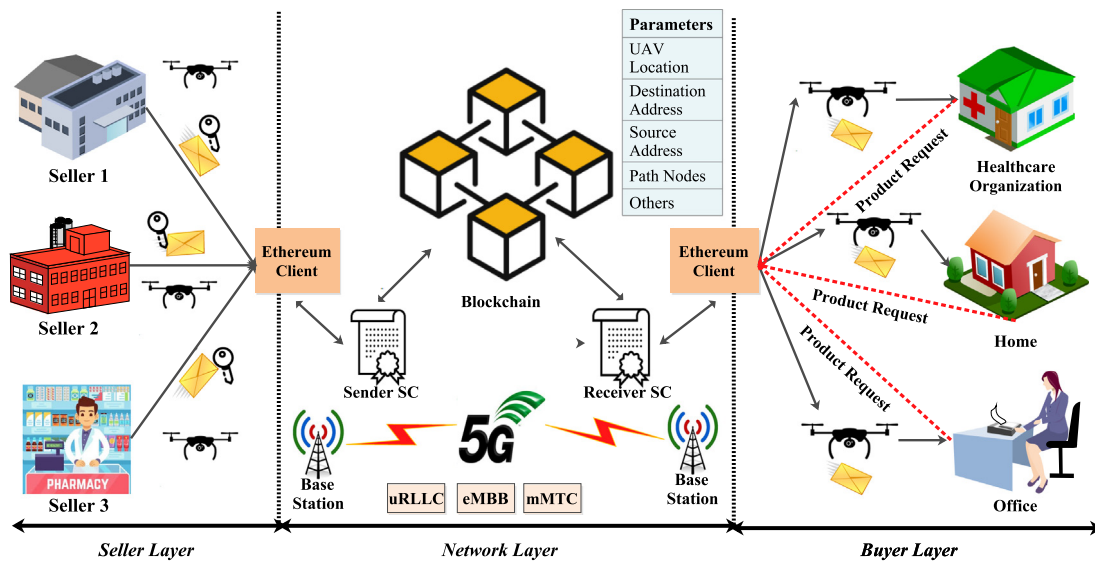


Fig. 15. Package delivery in blockchain-envisioned UAV network.

data in BC network is quite secure (records are immutable), reliable (a data is stored after is being verified by all participating nodes), and easily traceable (new update is appended into the block). The latency of the package delivery system is decreased with the involvement of 5G communication channel. Its uRLLC component can able to manage the latency to near <1 ms.

8. Conclusion

In this paper, we have presented detailed information on BC technology and UAV networks along with their security issues, vulnerabilities, and solutions. A discussion and comparison of various existing surveys are presented. A Taxonomy of various security issues of UAV related to 5G and network issues are discussed. Numerous features from various existing networks such as 5G are extracted (such as latency, reliability, and security) and implemented in UAV networks. But still, some issues remain unsolved in 5G-enabled UAV networks. This paper used distributed BC technology to solve the aforementioned issues. Then, we have listed some challenges being faced by the integration of BC in the UAV network. Finally, this paper presented a case study of package delivery in Industry 4.0 to explore the suitability of BC for UAV networks.

Declaration of competing interest

No author associated with this paper has disclosed any potential or pertinent conflicts which may be perceived to have impending conflict with this work. For full disclosure statements refer to <https://doi.org/10.1016/j.comcom.2020.01.023>.

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